

LTB Sales Portal - System Development Progress

Second Milestone

Currently, we are still performing analysis on the data as we obtain new sources or encounter uncertainties that can cause complexities to arise in the system. The goals of this milestone has been classified into preparing and exploring the data, and lastly modeling and evaluating the data. While the data modeling and evaluation performed has satisfied the purpose of this milestone, the data preparation and exploration will continue as new data, or uncertainties in the data, arrives. In fact, the complexity and uncertainty of the data have made this milestone somewhat challenging since we are still obtaining data that are essential for the system analysis and development.

Milestone Goals:

The goals of this milestone were based on first preparing and exploring the data, and secondly, modeling and evaluating the data.

1. The data analysis task was based on identifying the missing data sources, gathering and handling data, detecting and correcting corrupt or inaccurate data from the provided data sources. Note that a proper data preparation and cleaning cannot be performed until all necessary data is gathered. However, a good approximation can be estimated based on the system needs and requirements.
 - a. Data Preparation
 - b. Data Exploration
2. The second task, and main focus of this milestone, is modeling the system models based on the data provided. In order to meet the client's needs and requirements, the system has been designed as an integration of two models, the permissions-based system and the strategy-plan system. These two system models were the result of the implementation of the following phases:
 - a. Data Modeling
 - b. Data Evaluation

Project Progress:

A brief summary of the work done to achieve the goals mentioned above is being described below. However, to obtain a more descriptive information above the current state of the project you can access the files provided in the reference section of this report.

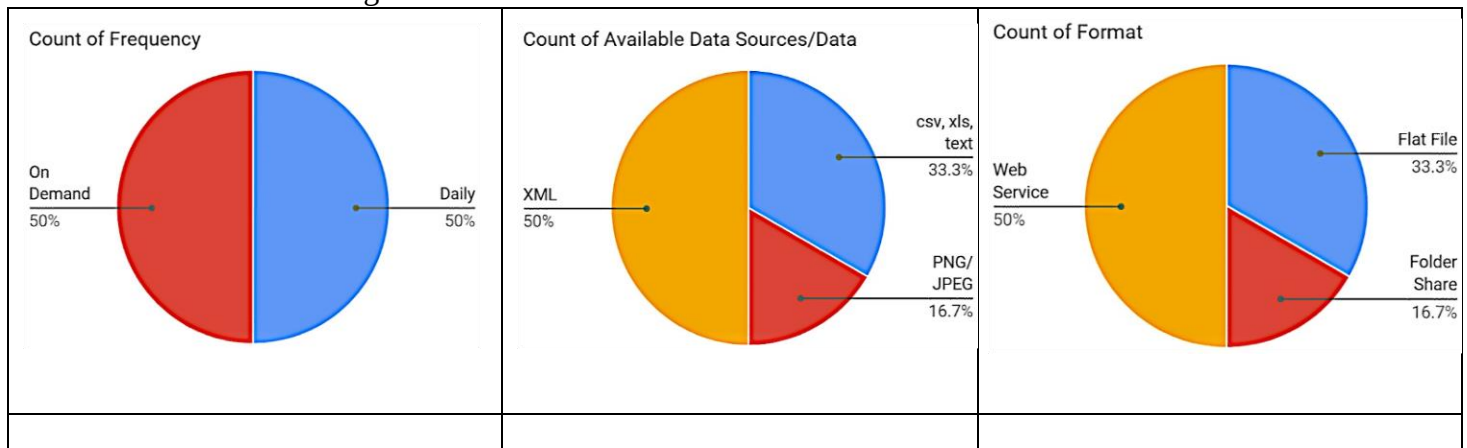
1. Exploring & Preparing the Data

The data is currently being explored and prepared as data sources are being provided or modified. Currently, a google spread sheet has been created to remotely keep track of the state of the data handling; this sheet can be access [here](#). The data analysis can be overwhelming, therefore, here we will only discuss the progress and results of the provided data. To obtain further information on the data analysis performed you can access the links such as [explore](#) and [analysis](#), which contain more details about the data. To efficiently explore and prepare the data, this has been categorized into available and pending data sources.

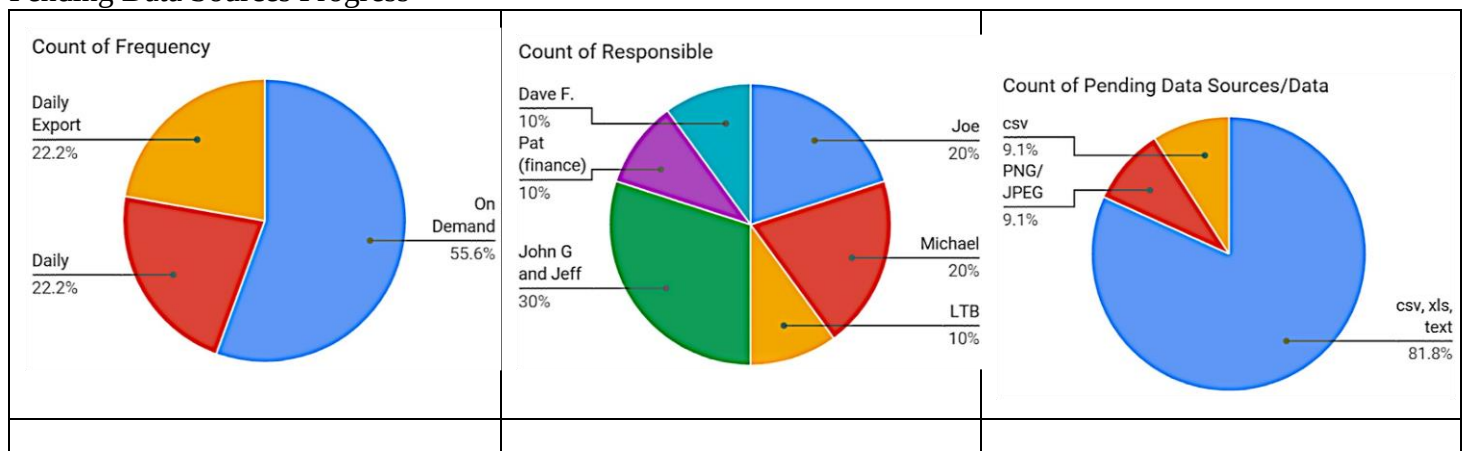
Documented by:
Piter Garcia

On behalf of:
True Interaction

Available Data Sources Progress



Pending Data Sources Progress



DataExploration.docx: document shows and describes the attributes of some of the data we are going to be handling for this project. Along with this document is a google spreadsheet (Data-Handling) that is being used to identify the data we have available and need to extract, or import, from, or to, a source, as well as the new sources that need to be built.

Files Location:

DataExploration.docx:

[Data Exploration](#)

Data-Handling.xls:

[Data Handling - Google Spread Sheet](#)

The folder DB Schema, contains everything that has been done about the architecture of the system. Other important files about this project can be access in the "LTB - Sales Portal" folder. These files are the features requested by LTB-Sales Portal, project overview, and the project outline.

LTB Sales Portal-Requested

Features.docx:

[Company Requested Features](#)

Project_Overview.docx:

[System Overview](#)

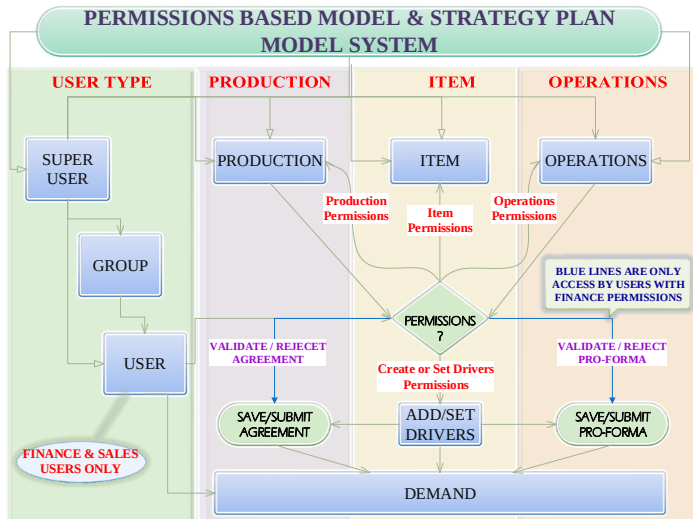
Project_Outline.docx:

[Project Outline](#)

The folder "Client Artifacts" contains every document provided by LTB-Sales Portal to us. [Clients Artifacts](#)

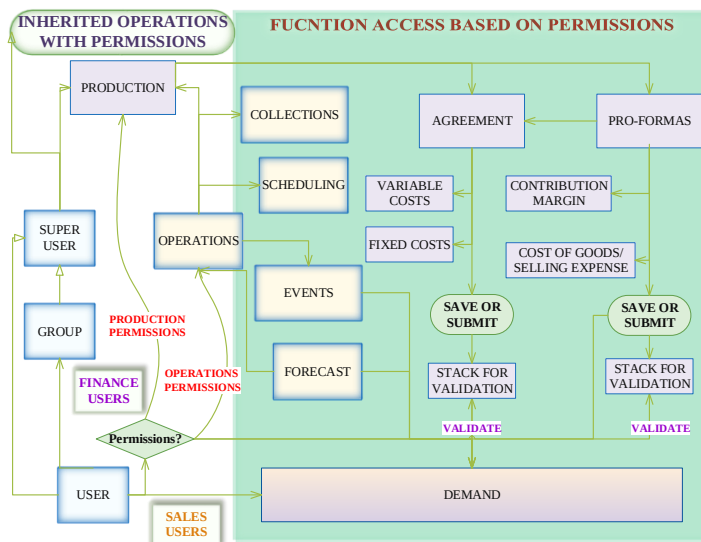
2. Evaluating and Modeling Data

Evaluating the data was an essential step to design the models, or structure, for the system architecture such as SCHEMA, UML and ERD.



Permissions-Based System

After evaluating the needs or requested features of the client, as well as the provided data, we decided to design a permissions based system to allow users, such as sales and finance agents and customers, to perform a series of tasks according to the group they belong to. To achieve this performance, the system must provide users access to functions that are inherited from the parent class. Users will be able to access these functions through the permissions of their group(s).



Strategy-Plan System

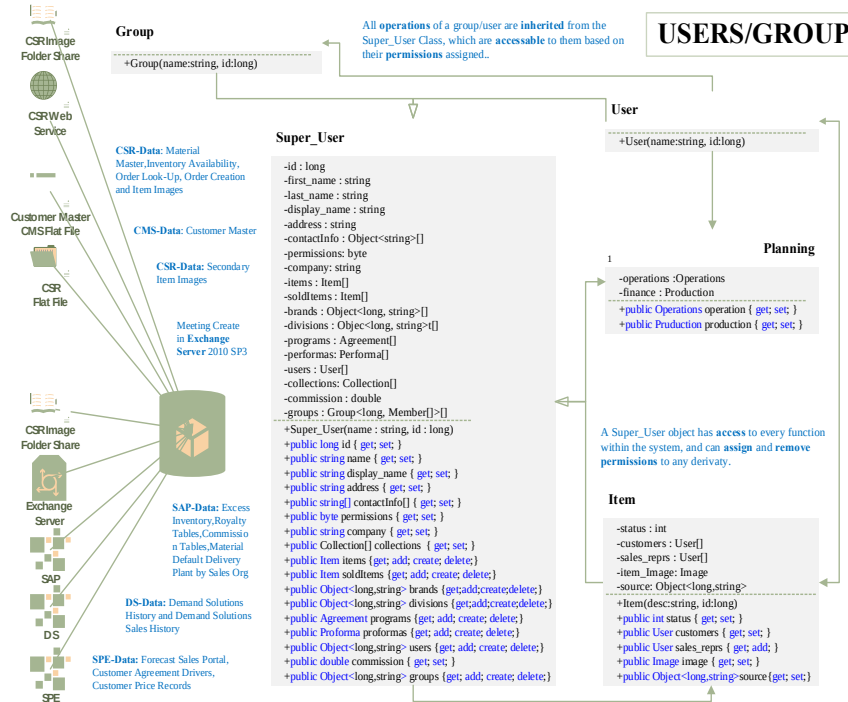
This system is mainly driven by the sections, operations and finance, respectively. Operations contains the forecast data, the collections object, and the sales plan, which contains the two most important objects that we will be working on; "agreements" and "proformas". Finance has access to production, which is also part of sales plan. Therefore, they also have access to agreements and proformas, which they will validate on a regular basis. Both, agreements and proformas must have a two validation process, the head of a division and a finance agent, in order to be active. Other tasks in the system are either part of both models integration, such as Demand Solution, or dependents such as schedule and

events.

GROUP/USER	Objects utilized to reference companies and groups, employees, customer and owners.
FINANCE	Provide permissions to a set of budgets that identify sources of funds for the planned operation and the uses for these funds during a period to carry out the budget activities.
OPERATIONS	Provide permissions to needed resources and the way that these resources will be acquired for all day-to-day activities such as sales and services, production, purchasing, marketing, research and development.
ITEM	Provide permissions to items and their basic information as well as their history and performance.
PERMISSIONS	Object utilized to privatize all functions in SUPER USER, the parent class, and provide users with permissions to access all, or some, of these functions according to the group they belong to.

Documented by:
Piter Garcia

On behalf of:
True Interaction



Designing Data Models

Permissions-Based System

The file **Permissions_Based_System** is a .docx file that contains a brief description of the system, as well as the objects and the data flow that we are more or less anticipating for the objects mentioned above. Along with this, is the **modelIntegrationSystem.vsd** "Visio Project" file in the "Project" folder, and in case you do not have access to this software you can access the pdf version in the "PDFs" folder.

Files Location:

Permissions_Based_System.docx:

[Permissions Based System](#)

ModelIntegrationSystem.vsd:

[Model Integration System - Visio](#)

ModelIntegrationSystem.pdf:

[Model Integration System - PDF](#)

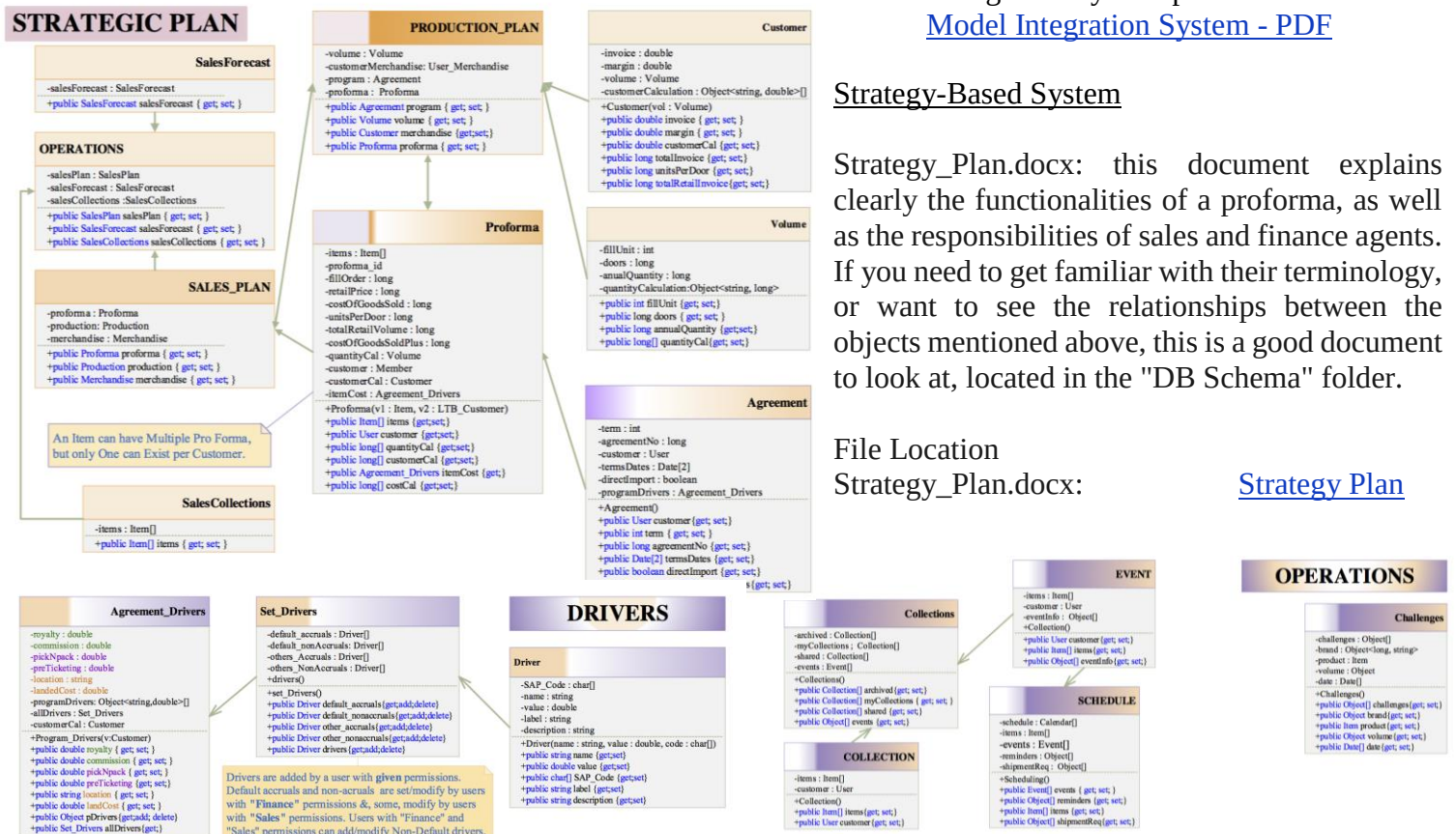
Strategy-Based System

Strategy_Plan.docx: this document explains clearly the functionalities of a proforma, as well as the responsibilities of sales and finance agents. If you need to get familiar with their terminology, or want to see the relationships between the objects mentioned above, this is a good document to look at, located in the "DB Schema" folder.

File Location

Strategy_Plan.docx:

[Strategy Plan](#)



Drivers are accrual, or non-accrual, values within an agreement that are used to drive the pricing of one or multiple items.

Operations contain plans that identify needed resources and the way that these resources will be acquired for all day-to-day activities such as sales and services, production, purchasing, marketing, research and development.

Documented by:

Piter Garcia

On behalf of:

True Interaction

SystemUI.docx: this document, located in "DB Schema", contains most of the wire frames designed by Liam and the data structure of its objects along the side. The structure given to these objects do not need to be exactly as the reference given in this document since the structure of each language is somewhat different. However, the relationship between objects, fields, members and functions and be used as a reference.

File Location:

SystemUI.docx:

[User Interface Data Structure](#)**Next Step:**

Right now our focus is data preparation. The plan is to have all data sources identified by tomorrow so we can quickly move to the next step in data preparation. Matt and I are meeting with Cliff tomorrow to close the gaps missing in the data handling document, the missing data sources, here is the link for you to review:

[Data Handling Google Spread Sheet](#)

We will be focusing on the following tasks:

- 1 - Data Preparation -> Data Acquisition, Scrub & Ingestion into DB
- 2 - Data Exploration -> Perform analysis on the data, find relationships and common patterns.
- 3 - Data Modeling -> Objects Ready, CRUD Controls per Object, and
- 4 - Evaluation -> Polling Schedule / Design Assets Ready / Data Export to SAP & Demand Solutions
(Export Function Manually)

Reference Files:**System and Data Structure Design****Week 0**

Sources

- Understanding the Company's Goal [LTB-Sales-Portal-Requested-Features](#)
- Understanding the Data [Understanding-Data-Directory](#)
- Design System Architecture [Data-Schema/PDFs/Model-Integration-System](#)
- Design Data Structure [Wires/Frames-for-User-Interface](#)

Data Analysis**Week 4**

- Data Preparation [Data-Handling-Google-Spread-Sheet](#)
- Data Exploration [Notes/Exploring-the-Data](#)
- Modeling and Evaluation [Notes/Exploring-the-Data/Training-and-Testing](#)